Federated Learning Framework using healthcare data(Type-2 Diabetes)

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1. Introduction

- **Overview:** Healthcare organizations generate vast amounts of patient data, but data privacy laws (HIPAA, GDPR) restrict data sharing.
- **Federated Learning (FL):** A novel approach that trains machine learning models across multiple decentralized hospital clients holding local data samples, without exchanging them.
- **Focus:** Predicting Type-2 Diabetes using the Pima Indians Diabetes Dataset while ensuring patient records never leave the hospital servers.
- **Key Concepts:** Localized training, model parameter aggregation, privacy preservation (Differential Privacy), and security protocols (SHA-512).

2. Literature Review

The framework builds upon foundational research in privacy-preserving AI:

- **Federated Averaging (FedAvg):** McMahan et al. (2017) demonstrated communication-efficient learning of deep networks from decentralized data.
- **Handling Heterogeneity (FedProx):** Li et al. (2020) highlighted ways to optimize FL in heterogeneous networks where data isn't uniformly distributed (non-IID).
- **Differential Privacy (DP):** Abadi et al. (2016) introduced robust mechanisms (gradient clipping and Gaussian noise) to further protect against model inversion attacks.
- **Medical AI Implications:** FL significantly lowers barriers for multi-institutional healthcare collaborations without compromising patient confidentiality.

3. Problem Statements and Objectives

Problem Statement

Traditional machine learning requires centralizing patient data onto a single server, which raises serious **privacy, regulatory, and ethical concerns**. Healthcare data is inherently siloed, imbalanced, and localized.

Objectives

1. **Develop a Privacy-Preserving System:** Implement a Federated Learning framework that only shares model parameter updates (weights) rather than raw records.
2. **Handle Data Heterogeneity:** Effectively train on highly imbalanced and non-IID datasets across hospital boundaries using FedProx.
3. **Ensure Security & Integrity:** Implement SHA-512 hashing to prevent model

4. Methodology

System Architecture

- **Clients (Hospitals):** Keep patient data local. Compute parameter updates using a custom Neural Network (DiabetesNet with BatchNorm & Dropout).
- **Central Server:** Aggregates gradients using Weighted FedAvg or FedProx and performs global evaluations.

Training Flow (Iterative Rounds)

1. **Broadcast:** Server sends global model weights to all participating clients.
2. **Local Training:** Clients train on local data (using Adam, Cosine LR, and weighted BCELoss).
3. **Secure Transmission:** Clients send updated weights (hashed via SHA-512) back to the server.

5. Requirement Analysis

Functional Requirements

- **Data Partitioning:** Ability to split datasets into simulated local environments (IID and non-IID / Dirichlet).
- **Distributed Training:** Support for isolated client-side training and central server aggregation.
- **Security Protocols:** Implementation of SHA-512 hashing for parameter integrity.
- **Evaluation Mechanism:** Server-side held-out dataset evaluation focusing on medical-grade metrics (AUC-ROC, Recall).

Non-Functional Requirements

- **Privacy Policy:** Zero raw data transmission (regulatory compliance).
- **Scalability:** System should easily scale to integrate new hospital clients.

6. SDLC Models Used

The project was developed incorporating phases from multiple Systems Development Life Cycle (SDLC) models:

- **Waterfall Model:**
  - Used in the initial phases for strict requirement gathering, defining the system architecture, mathematical formulas, and dataset identification.
- **Incremental Model:**
  - Applied during the implementation phase. First, standard FedAvg was integrated, followed by incremental additions like FedProx, SHA-512 validation, and Differential Privacy (DP).
- **RAD (Rapid Application Development) Model:**
  - Utilized for the machine learning prototyping phase. Allowed for rapid iteration, continuous testing of hyperparameters, and tweaking model architectures

7. Conclusion

- We successfully built a functional **Federated Learning Framework** for predicting Type-2 diabetes without centralizing sensitive patient data.
- **Performance:** Achieved competitive medical metric scores (AUC-ROC ~ 0.84) nearly matching standard centralized models.
- **Privacy & Security:** Integrated robust defenses including Differential Privacy and cryptographic integrity checks.
- **Impact:** Demonstrates that healthcare institutions can collaborate on advanced AI models while strictly adhering to data privacy and compliance laws.

8. Future Scope

1. **Secure Aggregation (SecAgg):** Implementing advanced cryptographic techniques to protect the central server from observing individual client updates, mitigating risks from malicious clients.
2. **Multi-modal Data Integration:** Expanding the framework to handle larger, complex medical datasets such as MRI scans or full Electronic Health Records (EHRs).
3. **Real-world Hospital Deployment:** Transitioning from simulated environments to interconnected, secure cloud deployments across actual medical institution IT networks.
4. **Adaptive Privacy Budgets:** Dynamically adjusting Differential Privacy noise relative to the risk evaluated at each client node.

9. References

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Thank You

Questions & Discussion